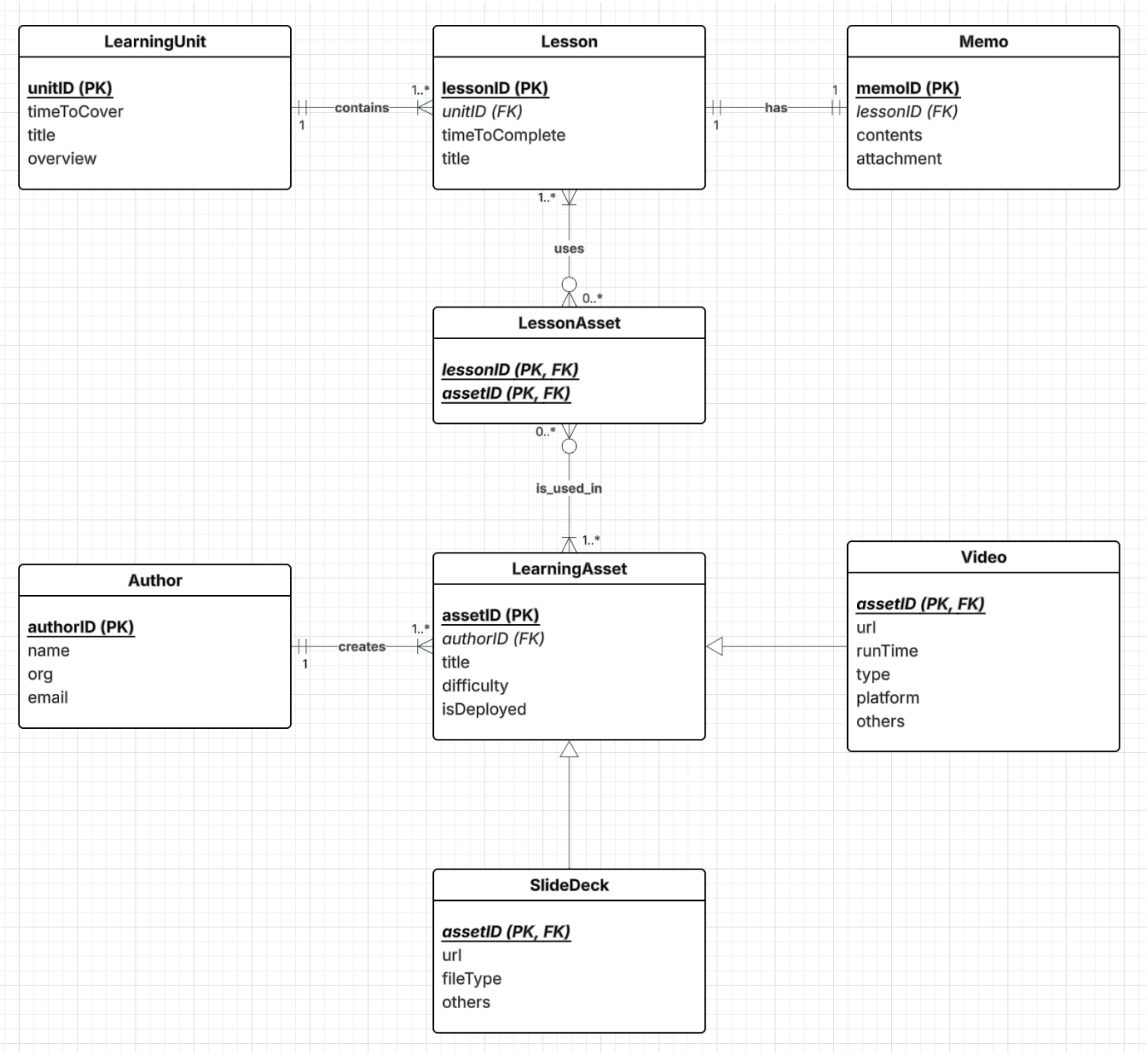
# Entity-Relationship Diagram (IE/Crow's Foot Notation)




**Legend:**

**| | = One (mandatory)**

**>| = One or more**

**>○ = Zero or more**

**Underlined = Primary Key**

**Italics = Foreign Key**

## Relational Schema Definitions:

**Author(<u>authorID</u>**, name, org, email)

**LearningUnit(<u>unitID</u>**, timeToCover, title, overview)

**Lesson(<u>lessonID</u>**, *unitID (FK)*, timeToComplete, title)

**Memo(<u>memoID</u>**, *lessonID (FK)*, contents, attachment)

**LearningAsset(<u>assetID</u>**, *authorID (FK)*, title, difficulty, isDeployed)

**SlideDeck(<u>*assetID (PK, FK)*</u>**, url, fileType, others)

**Video**(<u>***assetID (PK, FK)***</u>, url, runTime, type, platform, others)

**LessonAsset**(<u>***lessonID (PK, FK), assetID (PK, FK)***</u>)

## Relationship Explanations:

1. **Author → LearningAsset (1:M)**: One author creates one or more learning assets. Foreign key: authorID in LearningAsset

2. **LearningUnit → Lesson (1:M):** One learning unit contains one or more lessons. Foreign key: unitID in Lesson.

3. **Lesson → Memo (1:1):** Each lesson has exactly one memo. Foreign key: lessonID in Memo with unique constraint to enforce 1:1.

4. **Lesson ↔ LearningAsset (M:N):** A lesson uses 1..* assets; an asset can be used by 0..* lessons. Junction table: LessonAsset(lessonID, assetID) with composite PK (lessonID, assetID). This decomposes into two relationships:

   4a. **Lesson → LessonAsset (1:M)**: One lesson appears in one or more LessonAsset rows. Foreign key: lessonID in LessonAsset.

   4b. **LessonAsset → LearningAsset (M:1)**: Many LessonAsset rows reference one LearningAsset. Foreign key: assetID in LessonAsset.

5. **LearningAsset → SlideDeck/Video (Generalization):** Inheritance relationship. SlideDeck and Video reference LearningAsset.assetID as both PK and FK.

## Design Notes:

- All attributes are atomic (non-compound) and single-valued as required
- Primary keys ensure unique identification of all entities
- Foreign keys maintain referential integrity
- Many-to-many relationship (Lesson ↔ LearningAsset) implemented via LessonAsset junction table
- One-to-one relationship (Lesson → Memo) enforced via UNIQUE constraint on lessonID in Memo
- Generalization (LearningAsset → SlideDeck/Video) resolved using table-per-type inheritance with shared PK/FK
- Aggregation (LearningUnit contains Lesson) resolved into standard one-to-many relationship
- All relationships properly normalized and represented in Crow's Foot notation
- Enum types (difficulty, fileType, type, platform) can be implemented with CHECK constraints